2303A51502

Batch : 25

Task 1: Bubble Sort for Ranking Exam

Scores

Scenario

You are working on a college result processing system where a small

list of student scores needs to be sorted after every internal assessment.

Task Description

• Implement Bubble Sort in Python to sort a list of student scores.

• Use an AI tool to:

o Insert inline comments explaining key operations such as

comparisons, swaps, and iteration passes

o Identify early-termination conditions when the list

becomes sorted

o Provide a brief time complexity analysis

Expected Outcome

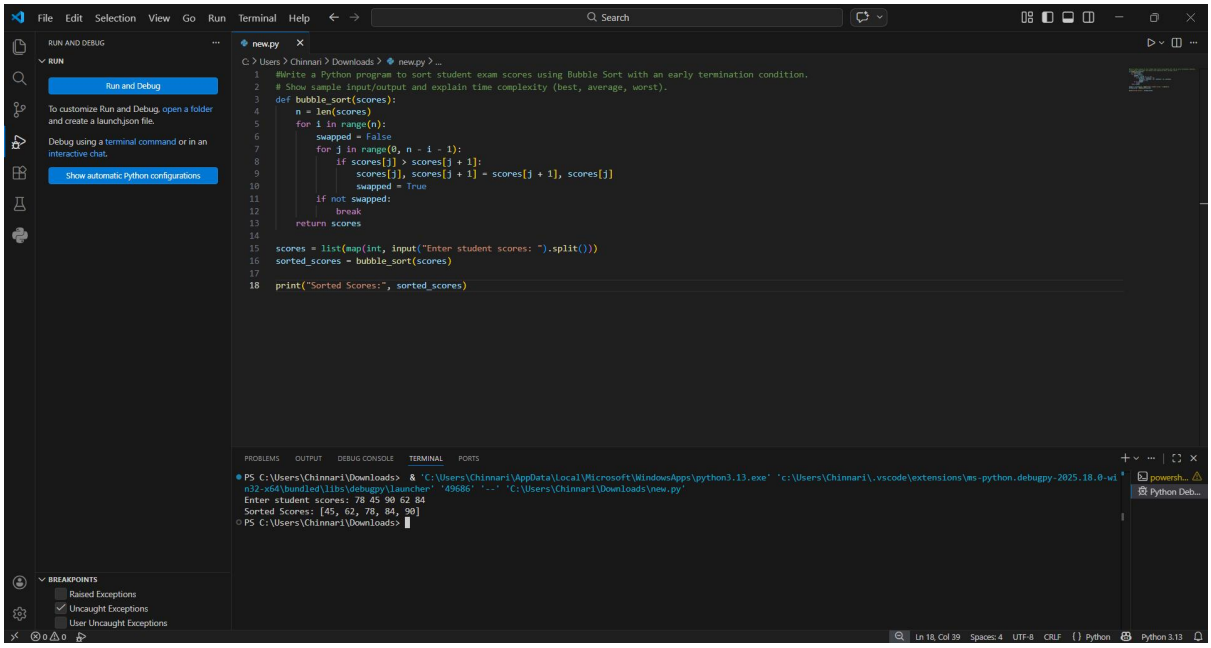
• A Bubble Sort implementation with:

o AI-generated comments explaining the logic

o Clear explanation of best, average, and worst-case

complexity

o Sample input/output showing sorted scores



Task 2: Improving Sorting for Nearly Sorted

Attendance Records

Scenario

You are maintaining an attendance system where student roll numbers

are already almost sorted, with only a few late updates.

Task Description

• Start with a Bubble Sort implementation.

• Ask AI to:

o Review the problem and suggest a more suitable sorting

algorithm

o Generate an Insertion Sort implementation

o Explain why Insertion Sort performs better on nearly

sorted data

• Compare execution behavior on nearly sorted input

Expected Outcome

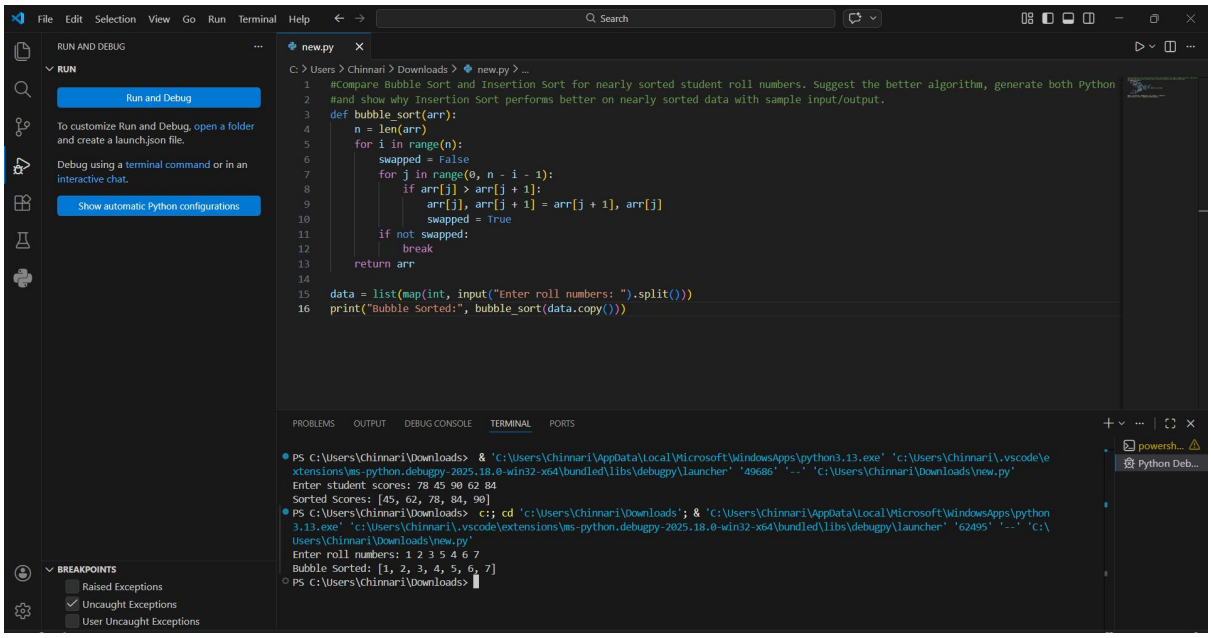
• Two sorting implementations:

o Bubble Sort

o Insertion Sort

• AI-assisted explanation highlighting efficiency differences for

partially sorted datasets give code and promt for this remove comments



Task 3: Searching Student Records in a

Database

Scenario

You are developing a student information portal where users search for

student records by roll number.

Task Description

• Implement:

o Linear Search for unsorted student data

o Binary Search for sorted student data

• Use AI to:

o Add docstrings explaining parameters and return values

o Explain when Binary Search is applicable

o Highlight performance differences between the two

searches

Expected Outcome

• Two working search implementations with docstrings

• AI-generated explanation of:

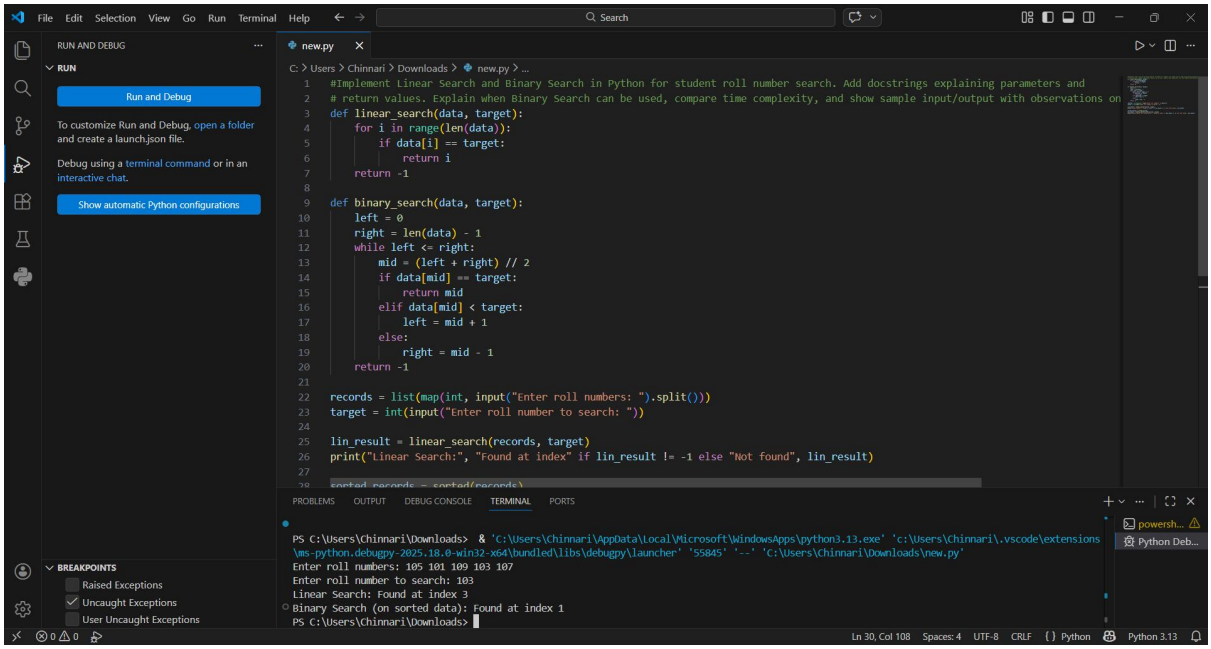
o Time complexity

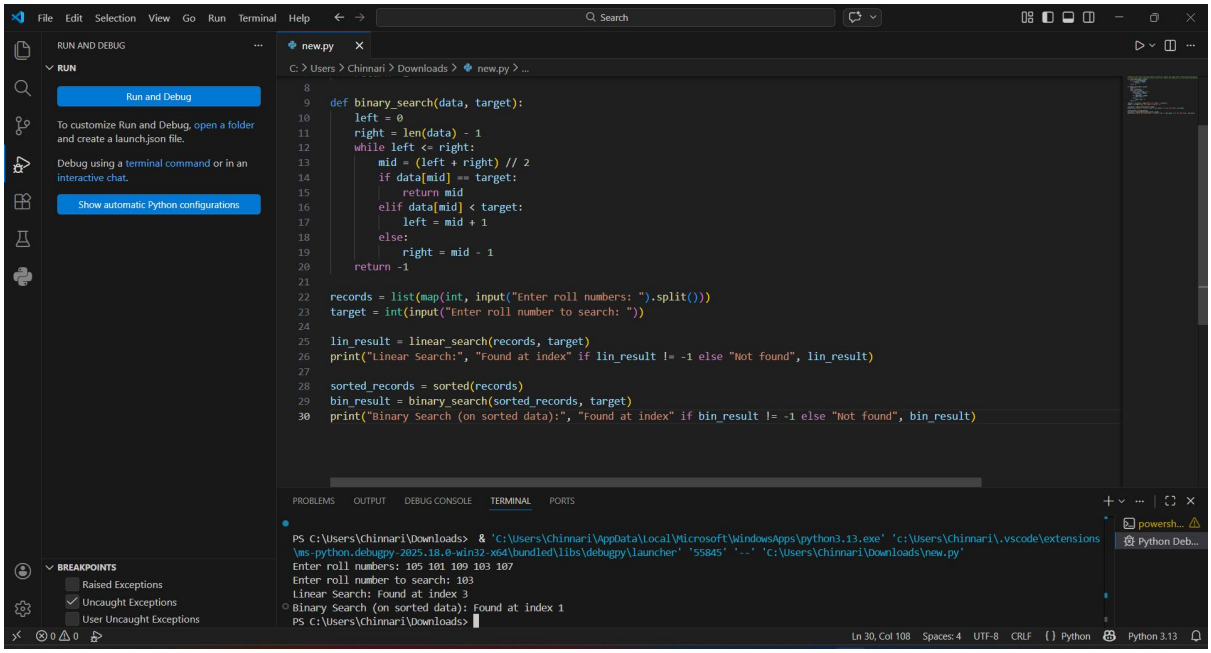
o Use cases for Linear vs Binary Search

• A short student observation comparing results on sorted vs

unsorted lists

give code and promt for this and remove comments





Task 4: Choosing Between Quick Sort and

Merge Sort for Data Processing

Scenario

You are part of a data analytics team that needs to sort large datasets received from different sources (random order, already sorted, and reverse sorted).

Task Description

• Provide AI with partially written recursive functions for:

o Quick Sort

o Merge Sort

• Ask AI to:

o Complete the recursive logic

o Add meaningful docstrings

o Explain how recursion works in each algorithm

• Test both algorithms on:

o Random data

o Sorted data

o Reverse-sorted data

Expected Outcome

• Fully functional Quick Sort and Merge Sort implementations

• AI-generated comparison covering:

o Best, average, and worst-case complexities

o Practical scenarios where one algorithm is preferred over the other give promt and code for this and remove comments

```python
#Complete recursive Quick Sort and Merge Sort functions in Python with docstrings. Explain recursion in both algorithms,
#test them on random, sorted, and reverse-sorted data, and compare time complexities and practical use cases.
def quick_sort(arr):
    if len(arr) <= 1:
        return arr
    pivot = arr[len(arr) // 2]
    left = [x for x in arr if x < pivot]
    middle = [x for x in arr if x == pivot]
    right = [x for x in arr if x > pivot]
    return quick_sort(left) + middle + quick_sort(right)

def merge_sort(arr):
    if len(arr) <= 1:
        return arr
    mid = len(arr) // 2
    left = merge_sort(arr[:mid])
    right = merge_sort(arr[mid:])
    return merge(left, right)
def merge(left, right):
    result = []
    i = j = 0
    while i < len(left) and j < len(right):
        if left[i] < right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
```

```
Enter numbers: 8 3 1 7 0 10 2
Quick Sort on Random Data: [0, 1, 2, 3, 7, 8, 10]
Merge Sort on Random Data: [0, 1, 2, 3, 7, 8, 10]
Quick Sort on Sorted Data: [0, 1, 2, 3, 7, 8, 10]
Merge Sort on Sorted Data: [0, 1, 2, 3, 7, 8, 10]
Quick Sort on Reverse Data: [0, 1, 2, 3, 7, 8, 10]
Merge Sort on Reverse Data: [0, 1, 2, 3, 7, 8, 10]
PS C:\Users\Chinnari\Downloads>
```

```python
def merge(left, right):
    while i < len(left) and j < len(right):
        if left[i] < right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1
    result.extend(left[i:])
    result.extend(right[j:])
    return result


data = list(map(int, input("Enter numbers: ").split()))

print("Quick Sort on Random Data:", quick_sort(data))
print("Merge Sort on Random Data:", merge_sort(data))

sorted_data = sorted(data)
reverse_data = sorted(data, reverse=True)

print("Quick Sort on Sorted Data:", quick_sort(sorted_data))
print("Merge Sort on Sorted Data:", merge_sort(sorted_data))

print("Quick Sort on Reverse Data:", quick_sort(reverse_data))
print("Merge Sort on Reverse Data:", merge_sort(reverse_data))
```

```
Enter numbers: 8 3 1 7 0 10 2
Quick Sort on Random Data: [0, 1, 2, 3, 7, 8, 10]
Merge Sort on Random Data: [0, 1, 2, 3, 7, 8, 10]
Quick Sort on Sorted Data: [0, 1, 2, 3, 7, 8, 10]
Merge Sort on Sorted Data: [0, 1, 2, 3, 7, 8, 10]
Quick Sort on Reverse Data: [0, 1, 2, 3, 7, 8, 10]
Merge Sort on Reverse Data: [0, 1, 2, 3, 7, 8, 10]
PS C:\Users\Chinnari\Downloads>
```

Task 5: Optimizing a Duplicate Detection

Algorithm

Scenario

You are building a data validation module that must detect duplicate

user IDs in a large dataset before importing it into a system.

Task Description

• Write a naive duplicate detection algorithm using nested loops.

• Use AI to:

o Analyze the time complexity

o Suggest an optimized approach using sets or dictionaries

o Rewrite the algorithm with improved efficiency

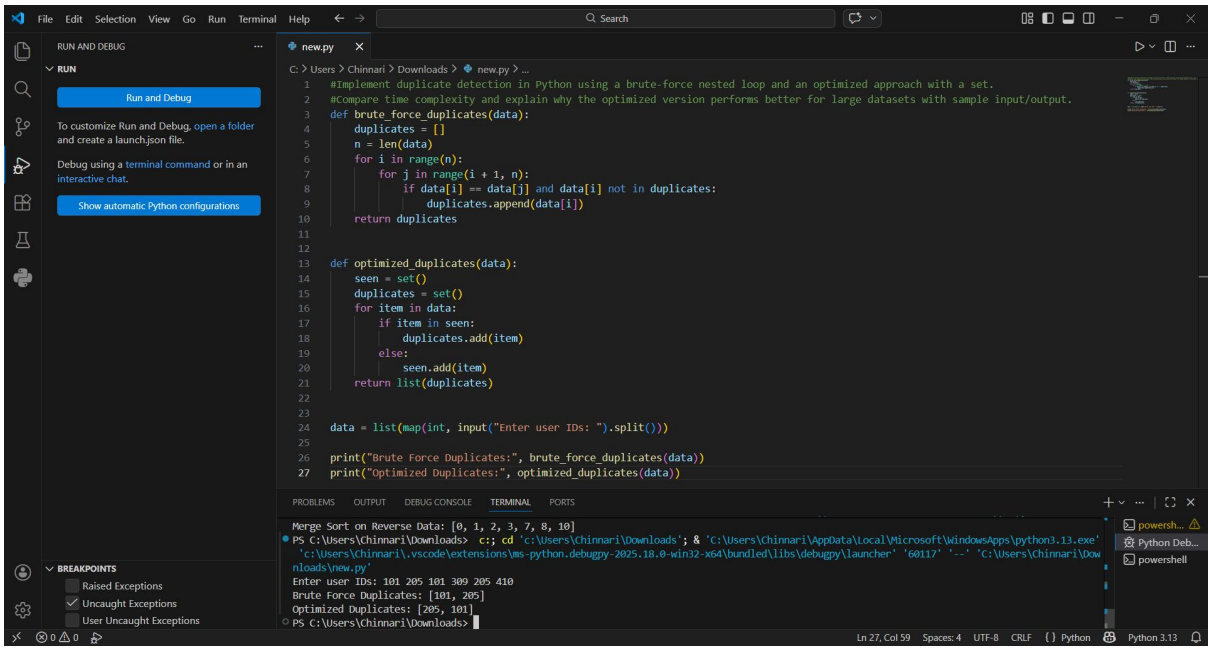
• Compare execution behavior conceptually for large input sizes

Expected Outcome

• Two versions of the algorithm:

o Brute-force ($O(n^2)$)

o Optimized ($O(n)$)

• AI-assisted explanation showing how and why performance

improved give promt and code for this and remove comments